6D STANDARD ENTHALPY CHANGE OF NEUTRALISATION

SPECIFICATION REFERENCE: 6.4 PART, 6.5, 6.8 PART

LEARNING OBJECTIVES

- Be able to define standard enthalpy change of neutralisation.
- Understand simple experiments to measure enthalpy changes in terms of evaluating sources of error and assumptions made in the experiments.
- Be able to process results from experiments to calculate a value for the enthalpy change of neutralisation.

WHAT IS MEANT BY STANDARD ENTHALPY CHANGE OF NEUTRALISATION?

The **standard enthalpy change of neutralisation** ($\Delta_{neut}H^{\ominus}$) is the enthalpy change measured at 100 kPa and a specified temperature, usually 298 K, when one mole of water is produced by the neutralisation of an acid with an alkali.

The following equation represents the standard enthalpy change of neutralisation of hydrochloric acid:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

For sulfuric acid, it is the enthalpy change for:

$$\tfrac{1}{2}H_2SO_4(aq) + NaOH(aq) \rightarrow \tfrac{1}{2}Na_2SO_4(aq) + H_2O(l)$$

For certain combinations of acid and alkali, the standard enthalpy change of neutralisation is remarkably constant. Some values are shown in **table A**.

ACID	ALKALI	$\Delta_{neut}H^{\ominus}$ / kJ mol^{-1}
HCl(aq)	NaOH(aq)	−57.9
HBr(aq)	KOH(aq)	−57.6
HNO_3(aq)	NaOH(aq)	−57.6

table A The standard enthalpy change of neutralisation is remarkably constant for certain combinations of acid and alkali.

All of the acids and alkalis in **table A** are classified as strong acids/alkalis. If we make the assumption that strong acids and alkalis are fully ionised in aqueous solution, then the reaction between them is simplified to:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

Because the reaction is essentially the same in each case, it is not surprising that the enthalpy changes are so similar.

EXPERIMENTAL DETERMINATION OF ENTHALPY CHANGE OF NEUTRALISATION

Here is how to find the enthalpy change of neutralisation.

- Wear safety glasses and a lab coat.
- Using a pipette fitted with a safety filler, place 25.0 cm^3 of 1.00 mol dm^{-3} acid into an expanded polystyrene cup.
- Measure the temperature of the acid.
- Using a pipette fitted with a safety filler, place 25.0 cm^3 of the alkali (usually dilute sodium hydroxide of a concentration slightly greater than 1.00 mol dm^{-3} (to make sure all the acid is neutralised)) into a beaker.
- Measure the temperature of the alkali.
- Add the alkali to the acid, stir with the thermometer and measure the maximum temperature reached.
- The appropriate laboratory apparatus is shown in **fig A**.

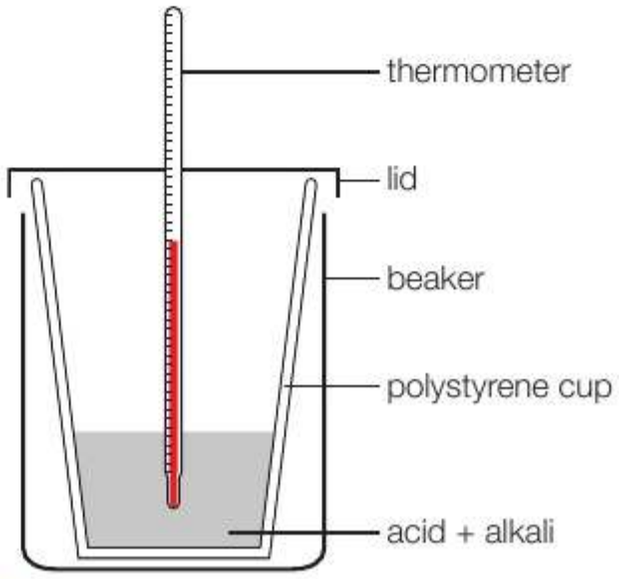


fig A Apparatus used to find the enthalpy change of neutralisation.

WORKED EXAMPLE

Find the enthalpy change of neutralisation.

Volume of 1.00 mol dm^{-3} HCl	25.0 cm^3
Volume of 1.2 mol dm^{-3} NaOH	25.0 cm^3
Initial temperature of acid	18.6 °C
Initial temperature of alkali	18.8 °C
Maximum temperature reached	25.4 °C (298.5 K)

Answer

Mean starting temperature = $\frac{1}{2}(18.6 + 18.8) = 18.7$ °C (291.7 K)

Temperature change, $\Delta T = (298.5 - 291.7) = +6.7$ K

Volume of solution heated = (25.0 + 25.0) = 50.0 cm^3

Mass of solution heated = 50.0 g (assuming that the density of each solution is 1.00 g cm^3)

Assume specific heat capacity of the solution = 4.18 J g^{-1} K^{-1}

$Q = 50.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times (+6.7\,K) = 1400.3\,J = 1.4003\,kJ$

Amount of acid neutralised = amount of water formed

= 1.00 mol dm^{-3} × 0.025 dm^3

= 0.0250 mol

$\Delta_{neut}H^{\ominus} = -1.4003\,kJ/0.0250\,mol = -56\,kJ\,mol^{-1}$

EVALUATING SOURCES OF ERROR AND ASSUMPTIONS MADE IN THE EXPERIMENTS

Answers should be given to only two significant figures. This is because ΔT is given to only two significant figures.

There are the usual uncertainties of measurements involved with the use of the pipette and the thermometer. Additionally, some heat energy will be transferred to the thermometer and the polystyrene cup.

CHECKPOINT

1. A **thermometric titration** was carried out using 1.00 mol dm^{-3} sodium hydroxide solution and dilute hydrochloric acid of unknown concentration.

50.0 cm^3 of the sodium hydroxide solution was placed in a suitable apparatus and 5.00 cm^3 portions of the hydrochloric acid were added. The mixture was stirred after each addition of acid and the temperature of the mixture was measured. Both solutions were initially at 20 °C.

The results obtained were as shown in the table.

VOLUME OF ACID / cm^3	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00
TEMPERATURE / °C	22.80	23.80	24.80	25.80	26.80	27.80	28.10	27.70	27.30	26.80

(a) Plot a graph of temperature change against volume of acid added. Determine the concentration, in mol dm^{-3}, of the hydrochloric acid.

SKILLS INTERPRETATION

(b) Calculate the enthalpy change of neutralisation, per mole of water formed, of the reaction. [Assume the specific heat capacity of the solution formed is 4.18 J g^{-1} K^{-1} and that its density is 1 g cm^{-3}.]

(c) Suggest suitable apparatus for carrying out the above experiment.

(d) In further experiments, the following enthalpy changes of neutralisation were determined, again using 1.00 mol dm^{-3} solutions at 20 °C.

HF(aq) and NaOH(aq) $\Delta_{neut}H = -68.6$ kJ mol^{-1}

CH_3COOH(aq) and NaOH(aq) $\Delta_{neut}H = -55.2$ kJ mol^{-1}

Suggest why each of these values is different from the values for HCl(aq) and NaOH(aq).

Suggest why these values are different from one another.

[Note: Part (d) of this question is beyond what is expected of you at IAS. However, if you do some research into what is meant by a weak acid, and also consider the energy changes involved when acid molecules ionise in water and also when the ions are subsequently hydrated, you should be able to answer it.]

2. In the experiment described in Question 1, a 50 cm^3 pipette was used to measure the volume of sodium hydroxide solution, and a 50 cm^3 burette was used to add the hydrochloric acid. The measurement uncertainty for the pipette is ±0.10 cm^3. The measurement uncertainty for each reading of the burette is ±0.05 cm^3 and for the thermometer is 0.05 °C.

(a) Calculate the total percentage measurement uncertainty in the use of:

(i) the pipette, (ii) the burette and (iii) the thermometer.

(b) Considering the description of how the experiment was carried out, suggest any possible procedural errors. State how you would carry out the experiment in order to minimise the effect of these errors.

SKILLS CREATIVITY

SUBJECT VOCABULARY

standard enthalpy change of neutralisation the enthalpy change measured at 100 kPa and a stated temperature, usually 298 K, when one mole of water is produced by the neutralisation of an acid with an alkali

thermometric titration a titration where the endpoint is indicated by a temperature change